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SUMMARY OF PROFESSIONAL ACTIVITY AT ISRAEL AEROSPACE INDUSTRIES (IAI)

Since joining IAI in December 2015, I have consistently developed and implemented advanced software solutions for real-time systems, research, and desktop applications, leading full development cycles from requirements definition through architectural design, implementation, testing, integration, and comprehensive technical documentation.

Airborne Software Department (December 2015 - October 2020)

During this period, I developed new control logic in C, from specification through implementation, working closely with avionics engineers on requirements definition and technical documentation. The logic primarily addressed engine, fuel, and fuel-pump management.

On my own initiative, I undertook an in-depth review and upgraded substantial sections of the UAV control software, which was difficult to maintain. My key activities included:

- Rebuilding and standardizing critical modules: digital inputs and outputs, serial links, nonvolatile memory, inter-computer communication (AVCs), and antenna management, among others.
- Improving performance on PowerPC processors and upgrading the development environment to newer versions of GHS Multi.
- Reorganizing the directory tree and arranging files by topic and device.
- Removing old, inactive, and redundant code; aligning coding styles; standardizing file and variable names; and improving readability and consistency while reducing coupling.
- Developing and integrating new capabilities to improve the reliability and safety of system operation.
- Developing a new avionics monitoring application (Avim) using C++ and Qt.

Algorithms Department (November 2020 - December 2021)

Developed a central management module in C++ on Windows for the Image Landing project, including requirements gathering, design, implementation, testing, and documentation.

Robotics Department - ELTA (January 2022 - June 2022)

Developed a central management module in C++ on Windows for the Worldview project, including managing internal communications among robotic subsystems.

Product Software Department (July 2022 - July 2023)

Conducted technology research and developed experimental applications in C, C++, C#, and Python.

Product Software Department (July 2023 - January 2024)

Developed a real-time production application in C++ for Windows for the GVDM project. The Israeli Air Force currently uses this software.

Product Software Department (February 2024 - December 2025)

Developed a signal-processing application in C++ for a UEI embedded system running Linux.

AI Software Department (January 2026 - Present)

Leading the development of an AI-powered application that translates complex aviation technical documents from Hebrew to English, including assembly instructions and inspection cards, while precisely preserving the structure and layout of the original PDFs. Designing a multi-stage quality-control framework in which independent verification agents automatically assess translations for structural, geometric, OCR, and semantic accuracy, recording every finding as evidence. This ensures that no unverified output reaches human review. Developing the product into a user-friendly Windows tool that allows non-engineer subject-matter experts to review findings, enter corrections, and build an increasingly refined glossary across hundreds of documents.

Mentoring and Collaboration

Over the years at IAI, I have also trained and mentored junior software engineers, collaborated with avionics engineers across technical disciplines, and helped others understand complex code and documentation.

Letters of Recommendation and Evaluations

I have numerous letters of recommendation and evaluation reports from clients, colleagues, and IAI managers that attest to my commitment, professionalism, and the broad impact of my work.

Kind regards,

Mikhail Jacques
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